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ORIGINAL PAPER

On the effect of bed condition on the development of tsunami-induced loading on structures using OpenFOAM

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Abstract In this study, a multiphase three-dimensional numerical model using the volume of fluid method is applied to investigate tsunami-like bores propagating over dry and wet flume beds and their interaction with a structural model. Physical results from a set of laboratory experiments conducted at the Canadian Hydraulics Centre of the National Research Council (NRC-CHC) in Ottawa, Canada, are used to perform a quantitative and qualitative validation of the numerical model results. Hydraulic bores, with varying initial downstream depths, generated by the sudden opening of a gated reservoir are released into a channel and impact a free-standing structure located downstream in the flume. Simultaneously, the authors analyze their propagation characteristics. Time-histories of run-up, pressure, and net base shear force acting on the structure placed in the downstream flume section are analyzed to further understand the development of hydrodynamic loading. Furthermore, an analysis of the velocity fields, before and during interaction with the structure, is presented to elucidate how the bed condition (wet or dry) effects water surface elevation and loading on the structural model.

Keywords Hydrodynamic loading · Tsunami · Air entrainment · OpenFOAM · Hydraulic bore

1 Introduction

On March 11, 2011 at 2:46 PM, a M_w 9.0 subduction earthquake occurred approximately 100 km off the Sanriku Coast, Japan (USGS 2013), generating a tsunami of unprecedented

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magnitude and spatial extent. More than 1,000 km of Japanese shoreline, ranging from the Chiba Prefecture in the south to the Hokkaido Prefecture in the north was directly affected by the ensuing tsunami wave. The broad low-lying coastal plains in the Tohoku (northeast) region were particularly devastated where the offshore bathymetry transformed the incoming tsunami waves into a series of turbulent hydraulic bores, which inundated some coastal regions by up to 10 meters water depth and reaching as far as 9 km inland (Miyamoto 2011). Estimates have put the death toll at nearly 20,000 and economic losses at over US$309 billion (USGS 2013), making it the most costly natural disaster ever recorded. Places as far as Hawaii and California reported damage to dozens of harbors, a testimony to the vast distances over which the destructive tsunami energy can travel, as well as raising a red flag to the potential danger faced by other tsunami-prone areas.

Post-tsunami forensic engineering reconnaissance missions, some of which the co-author has been involved in, have revealed that some structures previously understood to be invulnerable to tsunamis were heavily damaged or even destroyed during such events (Nistor et al. 2005; Ghobara et al. 2006; Yeh et al. 2013). In addition to these observations, an extensive literature review on current tsunami design documents indicated that hydrodynamic loading during tsunami inundation is not properly considered during the design of near-shore structures (Nistor et al. 2009).

The solitary wave and the dam-break wave have generally been used to replicate the coastal inundation by tsunamis in past physical modeling investigations. The solitary wave paradigm for tsunami modeling begins to appear following the milestone work of Hammack (1973) on the generation and propagation of tsunamis. In his dissertation, it was demonstrated that a positive initial surface displacement will eventually lead to the formation of leading solitary waves. However, Madsen et al. (2008) argued that it is often taken for granted that the geophysical scales provide sufficient propagation time for the formation of leading solitons. In addition, the authors point out that it can be misleading to focus on the leading short waves without addressing the main long wave on which they are riding. On the other hand, Chanson (2006) performed a frame-by-frame video analysis of a tsunami surge resulting from the 2004 Indian Ocean Tsunami advancing over dry coastal plains in Banda Aceh, Thailand. Equations developed to describe dam-break flows were applied to the surge data which revealed many similarities with respect to the surge profile and wave celerity, providing a strong physical analogy between the two phenomena.

In Ramsden's (1993) experiment, wet bed (bore) and dry bed (surge) dam-break waves were generated by the sudden release of an impounded volume of water which subsequently impinged on an instrumented vertical wall, extending across the entire flume width. Before impact, it was observed that the bores' leading edge was three times steeper than that of the surge and that at any point near the wave-tip, the bores' profile was significantly deeper. The impulsive or shock pressure recorded at the beginning of the time-history in Fig. 1b, especially in the case of the bore, has also been observed in previous studies such as Nakamura and Tsuchiya (1973). Although the magnitude of the impulsive pressure in the case of the bore is greater than that of the surge, Ramsden acknowledged that this result is rather inconclusive as measurements were not taken close to the bottom of the wall where the pressures are highest. In addition, the sampling rate of his transducers may have been too low to capture such transient fluctuations. In Fig. 1c, it can be seen that the maximum force exerted by the bore overshoots the subsequent hydrostatic force (after the bore has reflected and velocities tend to zero) by nearly 50 %, whereas the force exerted by the surge displays a more gradual increase up to the hydrostatic force without overshooting. It is interesting to note that this overshoot in force exhibited by the bore occurs just after the measured run-up returns to the quasi-steady level

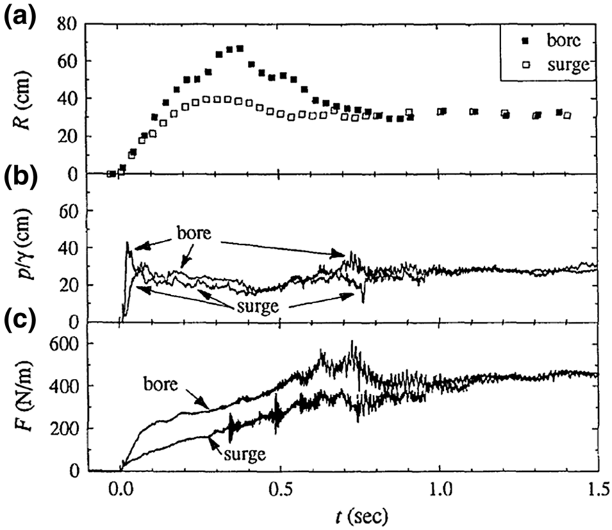


Fig. 1 Comparison of **a** run-up; **b** pressure head; and **c** force due to a turbulent bore and dry bed surge (Ramsden 1993)

after reflection of the bore. The same result was observed in the experiments of Cross (1967) who noticed that the maximum force occurred as the run-up tongue collapsed onto the incoming surge.

Using the dam-break wave approach, Árnason (2005) studied the interaction of bores propagating on a wet bed with surface-piercing structures of various cross-sections. Árnason's (2005) experiments were carried out in a flume 16 m long × 0.61 m wide × 0.45 m deep using a downstream depth of $h_{\mathrm{d}} = 0.02$ m and an initial upstream water level ranging from $h_{\mathrm{u}} = 0.1$ to 0.3 m. The force time-history generated by various initial impoundments using a square column with sides oriented parallel to the flume walls is shown in Fig. 2.

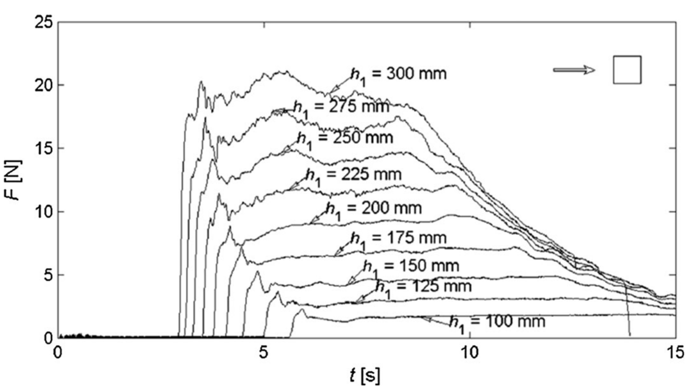


Fig. 2 Experimental time-history of stream-wise force acting on a *square column* for various initial impoundment depths (reprinted with permission from Árnason 2005)

It can be seen that for all but the smallest bores, the impulsive force does not overshoot the later quasi-steady hydrodynamic force. This result appears to be in contrast with the findings of Ramsden (1993), although the difference may be owing to the two-dimensional nature of Ramsden's experiment.

In an effort to better understand tsunami-induced hydraulic bores propagating over dry and wet channel beds and their interaction with surface-piercing structures, a numerical investigation of a recent physical experiment is conducted using a two-phase (air and water) three-dimensional numerical model within the framework of OpenFOAM. First, a description of the experimental test program and numerical model is given. The model is then validated in a qualitative and quantitative comparison of the numerical and physical results. Following validation, the model is applied to investigate the influence of the bed condition (dry or wet) on bore characteristics and how they affect the loading process on a structural model.

2 Experimental facilities

The physical model test runs were conducted in a 13.17-m-long × 2.70-m-wide × 1.4-m-deep high-discharge flume (HDF) at the NRC-CHC laboratory facilities, Ottawa, Canada (see Al-Faesly et al. 2012). The stainless steel-constructed flume was partitioned lengthwise using a 1/4″- thick sheet metal wall, leaving a 2.32 m opening at the upstream end. The partition subdivided the 5.6-m-long reservoir into two sections, effectively doubling the length of the flow path and thus increasing the duration of the hydraulic bores. An illustrated plan view of the setup is shown in Fig. 3. The bores were released into the remaining 7.57-m-long × 1.3-m-wide channel section and were generated by releasing an impounded volume of water using a rapidly swinging gate mechanism. Upon reaching the downstream end, the water was evacuated through a drain in the floor leading to a supply reservoir underneath the testing facility. Rapid gate openings (maximum of 0.3 s) were achieved by employing an electric winch and a counterweight system which, for all selected impoundment depths (*hu*) 0.55, 0.85, and 1.15 m, satisfied the ideal dam-break criterion defined by Lauber and Hager (1998).

The structural model used in the testing program was a square column, constructed out of four 6.3-mm-thick acrylic sheets, standing 1 m high with side dimensions of 0.3 m. An aluminum frame was fabricated to couple the structural model to a six-degrees-of-freedom (6DOF) load cell [MC6 Series, Advanced Mechanical Technology Inc.], bolted to the flume bottom. The aluminum frame was sufficiently rigid to ensure the effective and full transfer of the bore-induced loading to the measuring device. The upstream-facing side of the column was outfitted with ten differential pressure transducers [26PC Series,

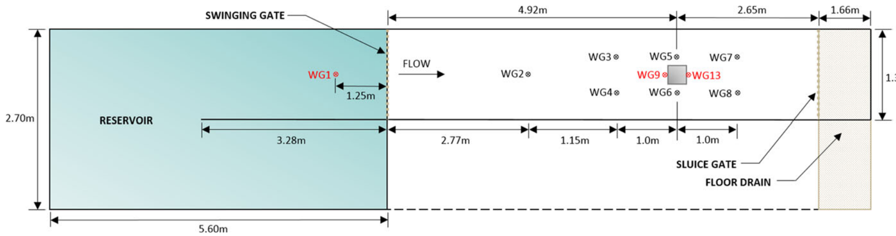


Fig. 3 Plan view of the physical model—wave gauge locations during runs with no structural model are shown in black

Honeywell] to measure the time-history of the vertical pressure distribution. Due to physical limitations imposed by the size of the transducer, the lowest sensor was mounted 20 mm above the flume bottom, while the subsequent one was installed at 50 mm. The remaining pressure sensors were spaced vertically at 50 mm intervals. Capacitance-type wave gauges [Model WP-AO-V02–03, Akamina Technologies] were deployed to monitor the water surface elevation at various locations in the flume (denoted as WG in Fig. 3). The sluice gate located at the downstream end of the flume was used to control water levels during calibration of the wave gauges. Before each test run, the data acquisition system was synchronized with the electric winch and set to collect data at a sampling frequency of 1,000 Hz.

3 Numerical model

All numerical simulations were performed using OpenFOAM v2.3.0 (Open Field Operations And Manipulation), an open-source CFD-toolbox produced by OpenCFD Ltd (OpenFOAM 2014). OpenFOAM is a collection of libraries and solvers written in the object-oriented C++ language, designed to run on UNIX-type systems like Linux. The open-source code is written in a syntax closely resembling tensor notation and partial differential equations, making it highly amenable to further development and modification. Several useful features such as the availability of 80+ solvers for a wide range of problems in continuum mechanics, pre- and post-processing utilities, error checks, and choice of spatial and temporal discretization schemes have made it an attractive choice for numerically modeling both research and practical engineering problems. Furthermore, code parallelization is easily achieved by decomposing the domain into a number of subdomains (one for each processor), allowing single models to be run over multiple cores or workstations.

The choice of solver for the current study was interFoam, an incompressible multiphase solver employing the Volume of Fluid (VoF) and interface capturing methodology for simulating free surface flows. The solver uses finite volume discretization (FVM) where the solution domain is represented by a number of control volumes or cells with computational nodes at cell centroids. The governing transport equations are then discretized and both spatially and temporally integrated over the control volumes.

3.1 Governing equations

The motion of a fluid can be described by the Navier–Stokes equations. Based on how the user wishes to treat turbulent fluctuations in the flow, OpenFOAM offers several different approaches. In the current work, the unsteady Reynolds-Averaged Navier–Stokes (RANS) equations are used, which are ensemble averaged to maintain statistically unsteady components of the flow. Expressed in tensor notation and Cartesian coordinates while neglecting body forces, the RANS conservation equations of mass and momentum of an incompressible fluid are given as (Ferziger and Perić 2002, p. 293):

$$\frac{\partial(\rho \bar{u}_i)}{\partial x_i} = 0 \tag{1}$$

and

$$\frac{\partial(\rho \bar{u}_i)}{\partial t} + \frac{\partial}{\partial x_j}\left(\rho \bar{u}_i \bar{u}_j + \rho \overline{u_i' u_j'}\right) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_j} \tag{2}$$

where ρ = fluid density, $\bar{u}_i$ = mean velocity component, u_i' = fluctuating velocity component, p = fluid pressure, and $\bar{\tau}_{ij}$ are the mean viscous stress tensor components, calculated as:

$$\bar{\tau}_{ij} = \mu\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right) \tag{3}$$

where μ = dynamic viscosity of the fluid.

However, in the framework of VoF and free surface modeling, some additional treatment is required. For purely Eulerian models containing two distinct phases, a common approach is to treat both phases as a single fluid with rapidly varying properties (Kothe 1999). In using such an approach, a "marker" field variable is introduced to describe the distribution of the two phases throughout the domain. Material properties (such as ρ, μ and ν) then become a function of this evolving marker field variable. In VoF methods, it is commonly known as the indicator function or phase fraction. The phase fraction, α, indicates the proportion of a cell which is filled with fluid at a given time step. Hence, for cells which are completely water or completely air, the phase fraction $\alpha(x_i, t)$ is equal to either one or zero, respectively. Cells containing the interface have a phase fraction value of somewhere between one and zero. The phase fraction is then convectively transported with the computed velocity field as follows (Heyns et al. 2013):

$$\frac{\partial \alpha}{\partial t} + \frac{\partial}{\partial x_i}(u_i \alpha) + \frac{\partial}{\partial x_i}(u_c|_i \alpha(1-\alpha)) = 0 \tag{4}$$

where $u_c = u_l - u_g$ is the relative velocity between the liquid and gas phases. A critical issue in numerical simulations of free surface flows using VoF models is obtaining a sharp resolution of the interface. Artificial numerical diffusion of the phase fraction tends to smear the region between phases over a few mesh cells and is therefore particularly sensitive to mesh resolution. One way to remedy this issue is by adding an additional convective term to the transport equation, appearing in Eq. (4) as the third term on the left hand side, known as the artificial interface compression term. The term *compression* stems from its effect to shrink the smeared interfacial region, thus resulting in a sharper interface. For more details on the surface capturing method, the reader is referred to Heyns et al. (2013).

In order to close Eqs. (1) and (2), a turbulence model is introduced based on the Boussinesq assumption where the Reynolds stresses ($\rho\overline{u_i'u_j'}$) are modeled using mean flow gradients and a turbulent eddy viscosity (Ferziger and Perić 2002, p. 294):

$$-\rho\overline{u_i'u_j'} = \mu_t\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right) - \frac{2}{3}\rho\delta_{ij}k \tag{5}$$

The turbulent kinetic energy (k) and its dissipation rate (ε) are computed from the following transport equations (Ferziger and Perić 2002, p. 295, 296):

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho \bar{u}_j k)}{\partial x_j} = \frac{\partial}{\partial x_j}\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j} + P_{\mathrm{k}} - \rho\epsilon \tag{6}$$

$$\frac{\partial(\rho \epsilon)}{\partial t} + \frac{\partial(\rho \bar{u}_j \epsilon)}{\partial x_j} = C_{\epsilon 1} P_{\mathrm{k}}\frac{\epsilon}{k} - \rho C_{\epsilon 2}\frac{\epsilon^2}{k} + \frac{\partial}{\partial x_j}\left(\frac{\mu_t}{\sigma_\epsilon}\right)\frac{\partial \epsilon}{\partial x_j} \tag{7}$$

where the rate of production of turbulent kinetic energy by the mean flow (P_{k}) and the turbulent eddy viscosity (μ_{t}) are calculated as:

$$P_{\mathrm{k}} = \mu_{\mathrm{t}}\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right)\frac{\partial \bar{u}_i}{\partial x_j} \tag{8}$$

$$\mu_{\mathrm{t}} = \rho C_\mu \frac{k^2}{\epsilon} \tag{9}$$

The turbulence model based on Eqs. (6) and (7) is the standard $k - \epsilon$ model which has been widely used. The five coefficients appearing in Eqs. (6)–(9) have been assigned standard values of $C_{\epsilon 1} = 1.44$, $C_{\epsilon 2} = 1.92$, $C_\mu = 0.09$, $\sigma_k = 1.0$, $\sigma_\epsilon = 1.3$ in the source code.

3.2 Initialization and boundary conditions

At the initial time, all computational nodes are prescribed an initial value for each field variable. In the current study, the field variables being solved for are phase fraction (α), velocity (μ_{i}), pressure (p), turbulent kinetic energy (k), and turbulent kinetic energy dissipation rate (ε). For computational nodes in the reservoir up to the selected impoundment depth (h_{u}), a phase fraction of $\alpha = 1$ was assigned. For simulations with a wet bed condition, the thickness of the bottom layer of mesh cells were adjusted to equal the desired downstream depth (h_{d}) and prescribed an initial phase fraction of $\alpha = 1$. By default, the remaining cells are treated as air ($\alpha = 0$). In the problem at hand, motion does not begin until after removal of the gate so the initial velocity field was set to zero. The initial pressure field was also set to zero. A small but nonzero value was applied to the initial values for k and ε to ensure good convergence in the first iteration of the solution. A generalized visual of the initial configuration is shown in Fig. 4a.

At cell faces coinciding with solid impermeable boundaries, the convective flux, and thus pressure gradient, is zero and the velocity is set equal to the walls (no-slip condition). The floor drain that allowed the water to leave the flume upon reaching the downstream end of the flume was modeled as an outlet boundary with a specified velocity gradient of zero. Pressure at the outlet face is calculated as $p = -1/2\rho|u|^2$ such that, for any value u, the pressure becomes negative, essentially mimicking the effect of the floor drain. It is important to note that since the opening of the gate satisfied the "sudden" dam-break

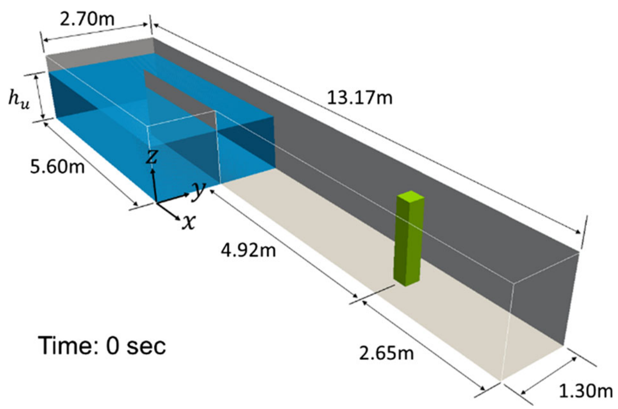


Fig. 4 Diagram of the initial physical model configuration

criteria given by Lauber and Hager (1998), it was decided that modeling the gate numerically would be redundant and therefore was not considered.

The computational domain was generated using a structured grid with an initially uniform distribution of $2 \times 2 \times 2$ cm grid cells. After generation of the mesh, internal cell faces coinciding with the partition in the reservoir were converted into an impermeable boundary with a no-slip velocity condition. A 1.5-m-long (5 column widths) section of the mesh extending across the flume and centered at the column location was refined to $1 \times 1 \times 1$ cm grid cells. In order to achieve maximum computational efficiency, a different mesh was generated for each impoundment depth ($h_u = 0.55, 0.85, 1.15$ m). The height of the mesh in the reservoir was set equal to the impoundment depth (h_u) plus one additional layer of grid cells to avoid adverse interactions observed between the liquid phase and boundary at the top of the domain. In the refined column region, the splash height at initial impact governed the height of the mesh which could reasonably be estimated from earlier runs with a coarser mesh. The height of the mesh in the channel section running between the reservoir and column region was governed by the height of an upstream-propagating wave induced by blockage effects in the flume.

3.3 Time step and solution procedure

The selection of an appropriate time step, especially in highly unsteady flows, can be difficult. A convenient way to work around this important issue was to make use of the adaptive time step control offered by interFoam. The use of an adjustable time step ensures accurate calculations as well as reducing the overall computational time, which was an important consideration. The time step is adjusted according to a prescribed maximum local Courant number ($\mathrm{Co}_{\max}$) and maximum allowable time step ($\Delta t_{\max}$) (Berberović 2010):

$$\Delta \mathrm{t} = \min\left\{ \min\left[\min\left(\frac{\mathrm{Co}_{\max}}{\mathrm{Co}_0} \Delta t_0, \left(1 + \lambda_1 \frac{\mathrm{Co}_{\max}}{\mathrm{Co}_0}\right) \Delta t_0 \right), \lambda_2 \Delta t_0 \right], \Delta t_{\max} \right\} \tag{10}$$

where $\lambda_1 = 0.1$ and $\lambda = 1.2$ are damping factors deployed to prevent oscillations in the time steps and Co_0 is the local Courant number. At the beginning of the calculation, the time step size is computed as:

$$\Delta t_0 = \min\left(\frac{\mathrm{Co}_{\max} \Delta t_{\mathrm{init}}}{\mathrm{Co}_0}, \Delta t_{\max} \right) \tag{11}$$

which is then used in Eq. (10) to ensure that the local Courant number (Co_0) at the initial time step is close to the prescribed limiting value $\mathrm{Co}_{\max}$.

The solution procedure to obtain the pressure field from the momentum equation (Eq. 2) utilizes the PISO (Pressure Implicit with Splitting of Operators) algorithm developed by Issa (1986). Here, at each time step, the velocity and pressure fields are solved in three steps:

- Momentum predictor step, which uses the last known solution for pressure to construct a first approximation of the velocity field by solving the momentum equation (Eq. 2)
- Pressure solution step, where the approximation of the velocity field is used to obtain an estimation for the pressure field
- Explicit velocity correction step, where the new estimate of pressure is used to compute new conservative mass fluxes and obtain new velocities

Since the velocity at one point is not only dependent on the pressure field but also the updated velocity field in its surroundings, the last two steps are iterated until a prescribed convergence tolerance is met.

4 Comparison of numerical and physical results

4.1 Effect of mesh resolution

As part of the sensitivity analysis conducted by authors, various mesh resolutions were tested to quantify the effect on model results. Although many different cell sizes were tested, it was decided that any grid cell size greater than 2 × 2 × 2 cm did not adequately reproduce details in the column region such as the splash at initial impact. Since the VoF methodology employed does not entail direct tracking of the interface location, a standard approach of integrating computed values of phase fraction over a vertical line extending from the flume bottom to the upper boundary was used. In this way, a time-history of surface elevation could be constructed for any desired location in the test channel. Figure 5a compares the experimentally monitored and simulated water levels at WG9 (run-up on the column face) for three meshes of varying grid cell volume: uniformly distributed 2 × 2 × 2 cm cells, uniformly distributed 1.5 × 1.5 × 1.5 cm cells, and a mesh generated with an initially uniform distribution of 2 × 2 × 2 cm grid cells which was refined to 1 × 1 × 1 cm in the near-column region (as shown in Fig. 4b). The initial peak in the time-history of surface elevation represents the splash that occurs following impact of the rapidly advancing bore. As evidenced by Fig. 5a, the mesh resolution has a substantial influence at this stage, up until a certain threshold. In moving from the 2 × 2 × 2 cm to the 1.5 × 1.5 × 1.5 cm mesh, a significant increase in the computed surface elevation is observed, improving the agreement with the experimental data. However, the further refined mesh with 1 × 1 × 1 cm cells in the near-column region (2 × 2 × 2 cm refined) does not appear to have any additional influence on the splash height.

By integrating the computed pressures across all column faces at each time step, time-histories of the net stream-wise force acting on the column could be constructed and compared to experimental measurements captured by the 6-DOF dynamometer. The experimental force data in Fig. 5b display a distinct peak occurring at approximately

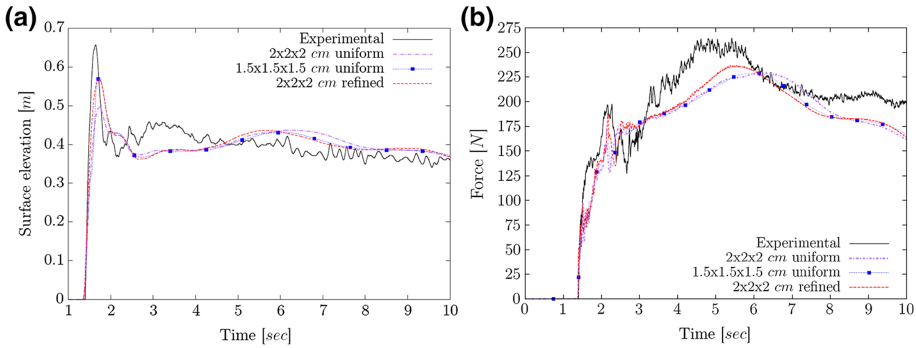


Fig. 5 Model sensitivity to changes in mesh resolution: **a** simulated column run-up (WG9) and **b** net stream-wise force on column for an initial impoundment depth, $h_u = 0.55$ m

$t = 2$ s, roughly 0.6 s after initial contact between the bore-front and column face. The numerical data display a similar trend. Upon further investigation, it was found that the timing of this peak coincided with the initial run-up tongue collapsing onto the incoming flow, an observation also noted in the experiments of Cross (1967) and Ramsden (1993). The three different mesh configurations show varying magnitudes of this initial peak. Although the experimental peak occurs over a longer period of time, the magnitude of the force appears to be reproduced accurately by the refined mesh with $1 \times 1 \times 1$ cm grid cells in the near-column region. Following the initial peak, the experimental and numerical data begin to diverge due to differing rates at which the force increases. A shift in the magnitude and occurrence time of the peak hydrodynamic force is observed when using the refined mesh, producing slightly better agreement. For this reason, subsequent simulations were performed using the refined mesh.

4.2 Model validation

A qualitative comparison of the computed and experimental water surface profiles for an impoundment depth of $h_u = 0.85$ m at different moments in time is given in Fig. 6. At $t = 1.9$ s, the initial run-up tongue has reached its peak elevation and begins to collapse under the influence of gravity. During this stage, hydrodynamic breakup of the surface is extensive and appears as spray reflecting off the upstream face and advecting around the sides of the column. Although the height of the simulated splash appears to be in fairly good agreement with the experimental observations, the hydrodynamic breakup of the surface is only moderately reproduced. This phenomenon is highly sensitive to the local mesh resolution and would require substantially smaller grid cells to increase the quality of the visual comparison. However, for the purposes of this study, this comparison is purely qualitative and therefore was not of primary consideration. In the second still-frame comparison, water surface profiles are compared at a later stage during the run. Strong flow acceleration around the column is evident by the rapid change in water elevation between upstream and downstream of the column. At a close inspection, a so-called rooster tail forming in the wake can be seen in its early stages in Fig. 6a, an observation that was also made during the experiments, although hidden behind the extensive spray as shown in Fig. 6b.

A simulation with an impoundment depth of $h_u = 1.15$ m was performed in the absence of a structural model. For validation purposes, numerical water levels are compared to the

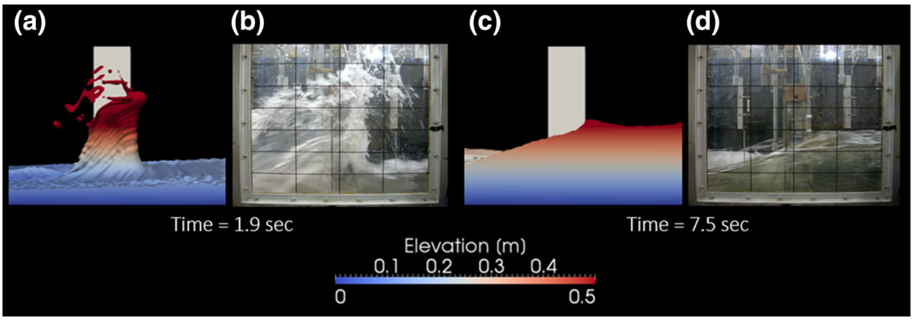


Fig. 6 Qualitative comparison of simulated (**a** and **c**) and experimental water surface profiles for $h_u = 0.85$ m

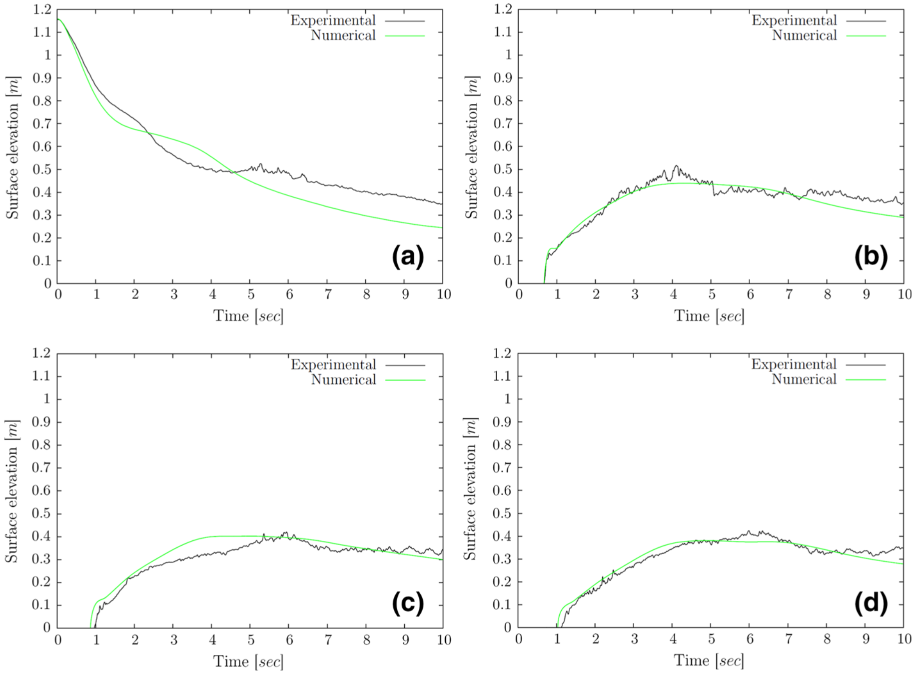


Fig. 7 Comparison of experimental and simulated water levels at **a** WG1; **b** WG2; **c** WG3 and **d** WG6 in the absence of a column. Initial impoundment depth $h_u = 1.15$ m

experimental data at wave gauges WG1, WG2, WG3, and WG6 which can be seen in Fig. 7a–d, respectively. In general, very good agreement is observed especially at WG2, WG3, and WG6. However, at WG1, some discrepancies begin to manifest early on when the water levels start to plateau at approximately $t = 1.5$ s. The numerical water level plateau's for a greater length of time, indicating that the back end of the numerical reservoir is recharging the front end at a faster rate than what occurred during the physical experiment. As a result, the simulated water level momentarily overtakes the experimental water level. The latter explanation would then help explain the discrepancy observed after $t = 4.5$ s. If the back end of the numerical reservoir was recharging the front end at a faster rate, then the water level at WG1 would, at some point, have to fall at a faster pace than the experimentally observed one due to a greater rate in reservoir depletion. Although this may be the case, the discrepancy may further be compounded by the fact that the computational domain did not consider an additional volume of water remaining in the large pipe that supplies the high-discharge flume. Despite these differences observed at WG1, water levels in the channel section appear to be reproduced with a high degree of accuracy, which for the purposes of this study was of greater concern.

4.3 Pressure and force

A comparison of experimental and computed time-histories of pressure, for an initially dry downstream channel bed, recorded and calculated, respectively, at 2 and 15 cm above the channel bottom is shown in Fig. 8. Fairly good agreement is observed. The most striking

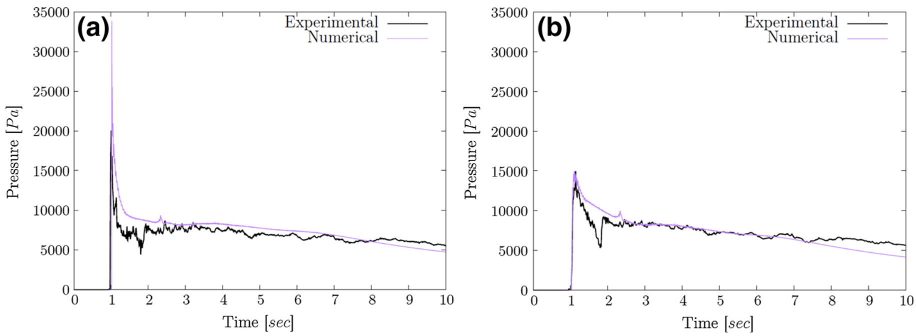
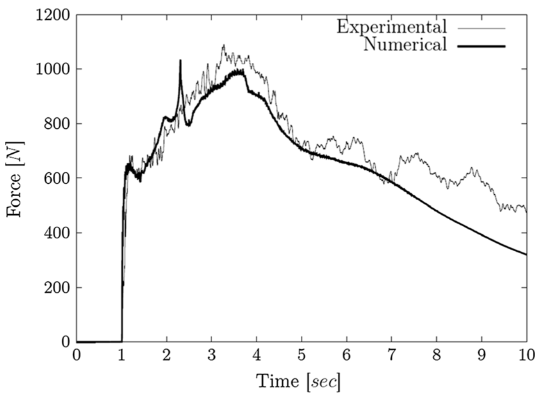


Fig. 8 Comparison of time-histories of the experimental and numerical pressure exerted on the upstream column face for the dry bed condition at **a** 2 cm and **b** 15 cm above the channel bottom. $h_u = 1.15$ m

discrepancy occurs at initial impact of the bore-front with the column which produces an impulsive pressure lasting less than a tenth of a second. This impulsive pressure is highest at the base of the column and decreases rather quickly in elevation. At 2 cm above the channel bottom, the model predicts a substantially higher impulsive pressure. Nevertheless, the discrepancy appears to reduce to nearly perfect agreement at higher elevations on the column face (see Fig. 8b). Despite the large differences between the computed and experimentally measured impulsive pressure at low elevations, the same discrepancy does not appear in the force time-history. This could perhaps be attributed to the difference being highly localized such that after integration of the pressure over the column face, it has a rather negligible influence. For about 1 s following the impulsive pressure, the numerical pressure tends to remain slightly above the measured value until reaching a quasi-steady state after $t = 2$ s.

The resultant stream-wise force acting on the column from integration of the computed pressures across the upstream and downstream column faces is displayed in Fig. 9. At the initial impact of the bore with the column, a small local peak in the force time-history occurs just after $t = 1$ s. In the study of St-Germain et al. (2012), the initial impulsive load generated at impact predicted by the single-phase smoothed particle hydrodynamics (SPH) model significantly overshot the experimentally observed value. At this stage of the loading process, due to the highly turbulent bore-front, the air entrainment has a large

Fig. 9 Comparison of time-histories of the experimental and numerical net base stream-wise force acting on the column for a dry bed condition. $h_u = 1.15$ m



influence on the magnitude of this impulsive load due to both increased compressibility and reduced water–air mixture density. Although even in the current model air is treated as incompressible, the reduction in fluid density could perhaps explain the reduced force at initial impact (compared to the SPH results), producing better agreement with the experimental data. Further on in the time-history, a sharp increase in the force occurs shortly after $t = 2$ s. As explained earlier, it was determined that the cause of this spike was the collapse of the initial run-up tongue onto the incoming surge. This pattern emerged for all simulated impoundment depths. However, it was noticed that in the experimental results that this effect was only visible for smaller impoundment depths (for example, see Fig. 5b). For the largest impoundment depth of $h_u = 1.15$ m, it is thought that the increase in pressure and force caused by the collapse of the run-up tongue is drowned out by other, more dominant effects contributing to the overall force. After approximately $t = 2.5$ s, the rate at which the force increases to the peak force in the experimental results (also referred to as "peak hydrodynamic" or "run-up" force) is in very good agreement.

The subsequent decrease of the net force is also predicted well by the numerical model. Similar to other time-history comparisons thus far, the numerical results tend to drop off faster during the last few seconds of simulation. As mentioned earlier, this discrepancy is caused by the additional volume of water not considered in the model that remained in the large pipe which supplies the high-discharge flume.

4.4 Effect of bed condition

Tsunamis occur as a series of waves which break near the shoreline before advancing inland as a turbulent bore. The first bore flowing overland will typically encounter dry coastal land (otherwise known as a "dry bed" condition). Depending on a variety of factors such as the magnitude of the tsunami, local topography, and wave period, subsequent tsunami waves may intrude inland before the inundating water of the preceding wave had time to fully recede (known as a "wet bed" condition). For such reasons, it is necessary to determine how the bed condition may influence bore propagation characteristics as well as how the latter will influence the magnitude of the hydrodynamic forces acting on coastal structures. A similar investigation by St-Germain et al. (2012) was performed using a single-phase SPH model. However, due to the repulsive forces induced by boundary particles in the model, there was substantial difficulty in producing desired downstream depths. As a result, only the smallest possible downstream depth of $h_d \approx 0.08$ m was tested. In the current study, any desirable downstream depth could be simulated by modifying the bottom layer of grid cells and setting appropriate initial conditions. An initial impoundment depth of $h_u = 1.15$ m was selected to investigate the effects of the bed condition. Herein, both dry and wet bed conditions of varying thickness $h_d = 0.005$, 0.02, and 0.05 m were simulated, and a qualitative comparison of the numerical results is further presented.

4.5 Bore characteristics

Analytical solutions for bore profiles obtained from the shallow water equations using the method of characteristics are compared to numerical bore profiles for bores propagating on dry and wet beds in Figs. 10 and 11, respectively. Ritter's (1892) dam-break solution for an inviscid fluid propagating on a frictionless dry horizontal bed and those later modified by Chanson (2009) are displayed in Fig. 10. Chanson's (2009) solution differs in the friction-dominated wave-tip region, described by the diffusive wave equation which includes a

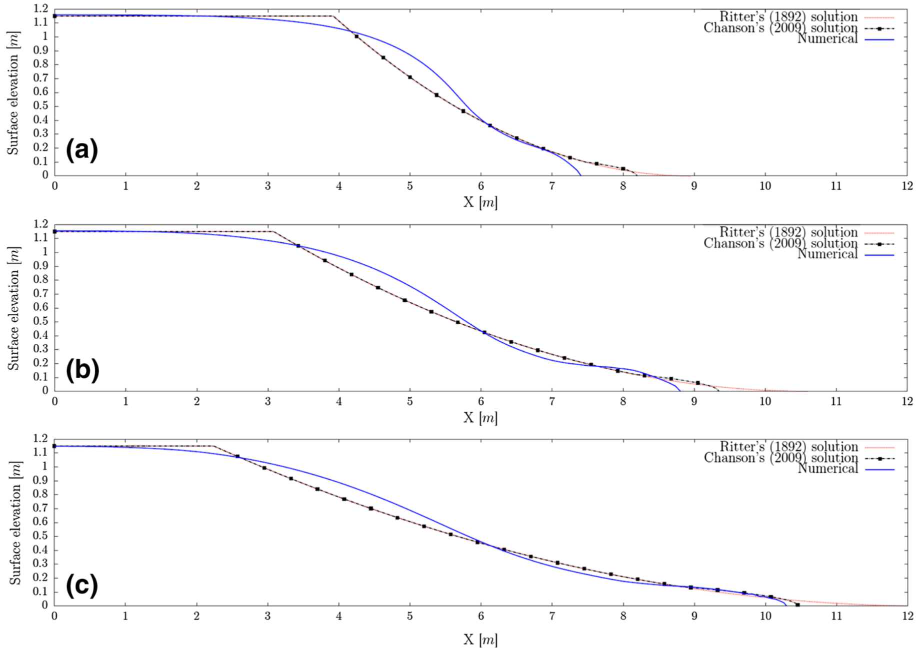


Fig. 10 Comparison of numerical and analytical solutions for bore profiles on a dry bed at **a** $t = 0.5$ s; **b** $t = 0.75$ s and **c** $t = 1$ s

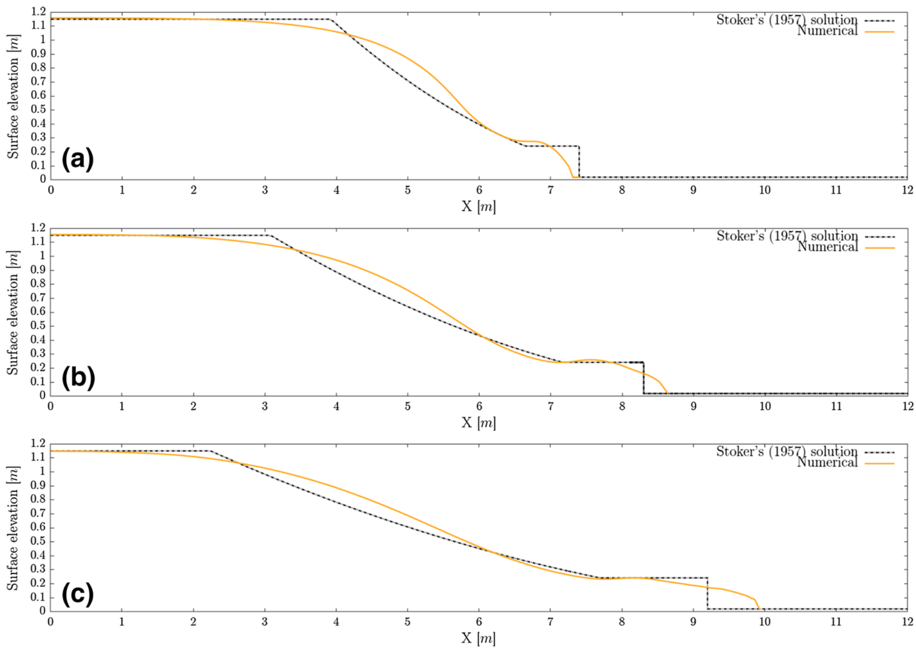


Fig. 11 Comparison of numerical and analytical solutions for bore profiles on a 20 mm wet bed at **a** $t = 0.5$ s; **b** $t = 0.75$ s and **c** $t = 1$ s

Darcy–Weisbach friction factor, f. Based on empirical curve-fitting data and descriptions of surface roughness, the value of f was taken as 0.03, which corresponds to a smooth surface, somewhere in between PVC and sand paper. Although the overall shape and wave-front celerity was relatively sensitive to the choice of f, fairly good agreement can be observed between Chanson's (2009) solution and the numerical bore profiles. This is especially true after the bore has had time to fully develop (e.g., $t = 1$ s in Fig. 10c). For bores propagating over a wet bed with an initial downstream depth of $h_d = 20$ mm, Stoker's (1957) analytical solution is used for comparison. Here, it can be seen that the depth of the wave-tip region is substantially larger than that in the case of a dry bed bore. The constant water level in the wave-tip region predicted by the analytical solution tends to approximate the depth of the bore-front relatively well. The greatest discrepancy appears in terms of the wave-front celerity where the constant $U = 3.63$ m/s predicted by the analytical model tends to underestimate the celerity of the numerical bore-front. In fact, in comparing the numerical bore-front locations for the dry and wet bed just prior to impact ($t = 1$ s), one can see that they only differ by a marginal amount. Although the shape of the bore was modified in a significant way by the presence of the initially quiescent downstream water, the bore did not appear to encounter substantial resistance. This is probably due to the relatively low downstream-to-upstream depth ratio of $h_d/h_u = 0.017$.

To further analyze the effects of bed condition on the propagation characteristics, a time-history of depth-averaged stream-wise velocity, measured 57-cm upstream of the column face, is provided in Fig. 12 for all tested initial downstream depths. After arrival of the bore-front, there is a steady and steep decline in the depth-averaged velocity until approximately $t = 6$ s, after which a transition to a much milder decline occurs. In addition, it appears that there are significant reductions in the bore-front celerity for increasing initial downstream depths. Upon further investigation, however, it was found that the presence of the initially still downstream water had a large impact on depth-averaged measurements. In Fig. 13, an elevation view of a vector plot for a bore propagating on a 50-mm wet bed is shown at three points in time before impact with the structural model. Shortly after removal of the gate, at $t = 0.25$ s, the initial jet of water that forms in response to the change in pressure gradient is directed at a slight upward angle due to the resistance imposed by the 50 mm of still water which acts like an inclined ramp. However, this upward redirection of the water is still observable up until impact and is visible by an undulation in the water surface behind the bore-front. The location of this undulation marks the beginning of a section of the bore-front that appears to ride on top of

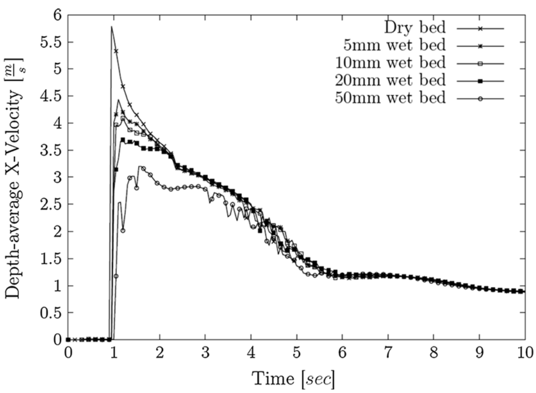


Fig. 12 Depth-averaged stream-wise velocity of bores propagating over different bed conditions ($h_u = 1.15$ m)

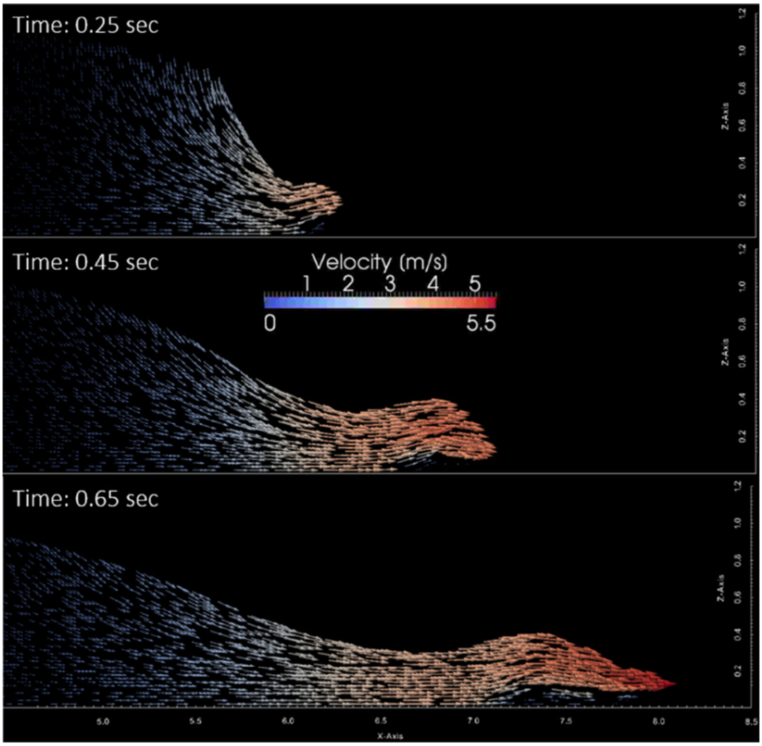


Fig. 13 Elevation view of the initial stages of bore development after gate removal ($h_u = 1.15$ m, $h_d = 0.05$ m)

the initially still downstream water. Behind the surface undulation, the released upstream water has had sufficient time to transfer its momentum to the underlying still layer such that the vertical velocity distribution appears to be much more uniform. The discontinuity in vertical velocity distribution in the bore-front region explains why such drastic reductions in the depth-averaged velocity for increasing downstream depths appear in Fig. 12. It can be seen in Fig. 13c that the tip of the bore has a velocity of about $U = 5.5$ m/s, which is just slightly lower than the wave-front celerity observed in the case of a dry bed. However, the tip region which exhibits the highest velocity is also the thinnest portion of the bore. As a result, the underlying still layer tends to have a large influence on the depth-averaged characteristics. Although the depth-averaged velocities may infer that there are significant increases in resistance to advancement of the bore for increasing downstream depths, the 50-mm wet bed bore arrived at the column location only 0.12 s later than the dry bed surge.

Comparisons of bore profiles for all tested bed conditions immediately prior to impact are displayed in Fig. 14. Since it was shown that the initial downstream water underneath the bore-front is essentially motionless (and therefore does not contribute to impulsive loading), the bore profiles are provided with respect to a reference elevation at the initial downstream water level. In this way, better comparison of the shape of the bore which actually exerts a dynamic pressure on the column can be made. Figure 14b shows that in general, the wet bed condition contributes to the accumulation of a larger volume of water

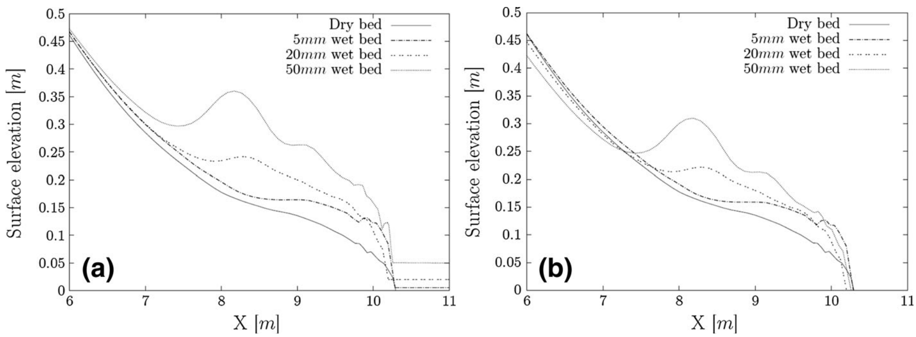


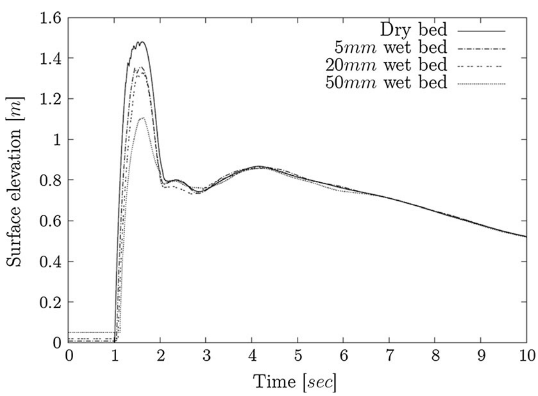
Fig. 14 Comparison of bore-front profiles for dry and wet beds of various depths immediately prior to impact with **a** reference elevation at flume bottom and **b** reference elevation at initial downstream water surface level

near the leading edge of the bore. As a result, the near-vertical face of the bore-front which impacts the column is extended vertically in the case of wet bed bores. However, with regard to the depth of the initial downstream water, only marginal differences can be noted at the leading edge. The most significant difference is the magnitude of the surface undulation which occurs upstream of the bore-front, and therefore does not contribute to impact dynamics.

4.6 Run-up

A comparison of run-up elevations on the upstream column face for the four tested bed conditions is provided in Fig. 15. The numerical results demonstrate that the height of the splash following initial impact is substantially influenced by the bed condition. The splash reaches the highest elevation for cases of bores propagating over dry channel beds, whereas a consistent reduction in the splash height is observed for increasing downstream depths. Between the dry bed and 50-mm wet bed, the two most extreme cases, a reduction of over 25 % in the peak splash height is observed. This is a rather surprising result in light of Ramsden's (1993) experiments during which he observed that peak splash height was increased for bores propagating over wet beds. In his discussion, the increase in splash

Fig. 15 Run-up elevation on the upstream column face for different bed conditions



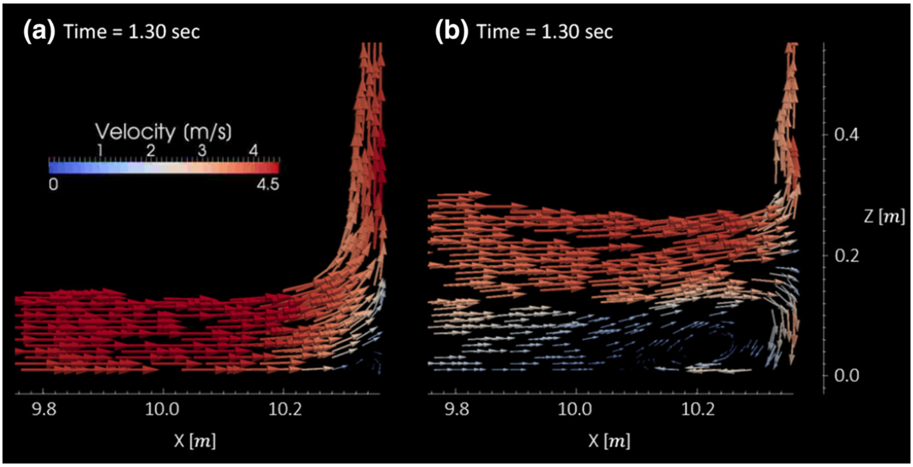


Fig. 16 Two-dimensional elevation view of the velocity field upstream of the column for **a** dry bed condition and **b** 50-mm wet bed condition just before initial splash reaches peak elevation

height was attributed to a greater volume of water being carried in the bore-front. However, it is worth noting that Ramsden's experiments involved the impact of bores on a vertical wall extending entirely across the flume, effectively reducing it to a two-dimensional problem. This may be an important distinction considering the three-dimensional flow around the structure in the present study. Interestingly, the bed condition appears to have no influence on run-up after collapse of the initial splash.

Intuitively, the splash height should be positively correlated with the wave-front celerity of the approaching bore. Given that the time of impact of the dry bed surge and 50-mm wet bed bore was 1.0 and 1.12 s, respectively, and a propagation distance of 4.77 m (distance between reservoir and upstream column face), the difference in average wave-front celerity was only 0.5 m/s. With such a large decrease in splash height, it seems unlikely that a change in wave-front celerity of 0.5 m/s is the sole factor at play. According to Ramsden (1993), the bore-tip volume also plays a role in splash height, but his positively correlated observation between volume and splash height are in contradiction with the current results. The only logical question left seems to be: where does the water go during impact? Figure 16 helps answer this question: as shown earlier in Fig. 13, at the time of impact, the front 2 m or more of the bore is riding on top of the initially quiescent water layer lying beneath. In Fig. 16b, a clear division can be seen between these two layers. As a result, the dynamic pressure exerted by the impacting bore is elevated above the channel bottom, centered at an elevation of approximately 0.2 m in the case of the 50-mm wet bed. Since this pressure does not extend all the way down to the bottom of the flume, it creates a divergence of the flow which is clearly visible in the velocity field. In fact, it appears as though only the top half of the impacting bore is diverted upwards, creating the splash. The other half appears to be diverted downwards which becomes part of a horseshoe vortex, beginning at the toe of the column which extends laterally and eventually wrapping around the sides of the column. Conversely, in the case of the dry bed, the dynamic pressure exerted at impact extends all the way to the channel bottom. As such, the generated pressure gradient drives the entire section of the bore upwards, creating a splash of larger magnitude.

4.7 Force

The resultant stream-wise force acting on the column is shown in Fig. 17 for all four tested bed conditions. It can be seen that after about $t = 4.5$ s the bed condition ceases to have any influence on the net force. However, several notable differences can be observed prior to this instant. First, the initial downstream depth appears to influence the impulsive load generated at initial impact with the structure. In the case of the bore propagating over a 5-mm wet bed, and only in this case, the magnitude of the impulsive load exceeds that generated by impact of the dry bed surge. The two larger downstream depths (20 and 50 mm) appear to have a damping effect at this stage of the loading process. As previously shown in Fig. 14, even the thinnest layer of water in the downstream channel section can have a substantial effect on the bore profile, increasing both the steepness and depth of the bore-front. However, further increases in the downstream depth beyond 5 mm did not seem to induce any significant changes near the bore-face. Since the steepness and depth of the bore-fronts for all wet bed conditions are similar, the small differences in the impulsive load must be due to a combination of bore-front celerity and air entrainment. Perhaps in the case of the 5-mm wet bed, the amplification of the impulsive load caused by an increase in the steepness of the wave-front overrides any damping effects caused by additional air entrainment or reductions in wave-front celerity. For further increases in initial downstream depth, the reduction in wave-front celerity and fluid density caused by air entrainment may become more important than wave-front steepness, resulting in a dampening of the impulsive load.

The second spike in the force time-history that occurs just after $t = 2$ s also appears to be influenced by the bed condition. In this case, it is the speed and volume of water which collapse onto the incoming surge which dictates the magnitude of this spike. Hence, the differences can easily be explained by the amount of time the run-up tongue has to accelerate on its downward trajectory. Since the depth of water just upstream of the column at the time of impact increases for increasing downstream depths (refer to Fig. 16), and the peak run-up height at initial impact reduces for increasing downstream depths (refer to Fig. 15), the distance the run-up tongue falls before colliding with the incoming surge also decreases. This explains the consistent reduction in this spike for increasing downstream depths.

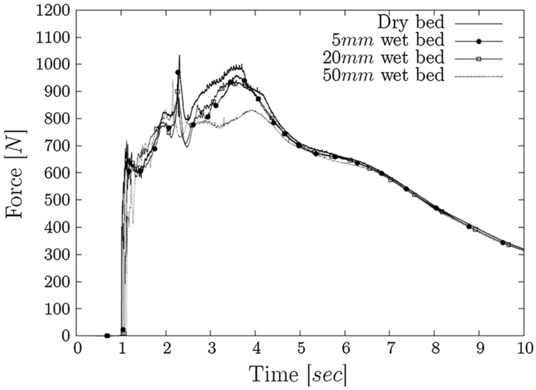


Fig. 17 Influence of bed condition on the time-history of the net stream-wise force acting on the column

The greatest and perhaps most important distinction appearing in Fig. 17 occurs during the peak hydrodynamic force at around $t = 3.5$ s (which was also the governing force in the physical results). Although a small reduction in the peak force can be observed at this time for the 5- and 20-mm wet bed in comparison to the dry bed, there appears to be a much more substantial decrease in the case of an initial downstream depth of 50 mm. Since it is known that water levels on the upstream face of column are not affected by the bed condition between seconds 3 and 4, any discrepancies in the differential hydrostatic pressure contributing to the net force must be caused by surface elevations in the wake. However, owing to a few instances of rapidly varying surface elevation at the leeside face of column, it was decided to show pressure acting at the bottom of the leeside face of the structure in Fig. 18 instead of the surface elevation. It can be seen that pressures acting at the leeside of the column, which are, for the most part, hydrostatic, begin to rise earlier for increasing downstream depths indicating a more rapid closing of the wake. In fact, the difference can be quite substantial. For example, at $t = 3$ s, the hydrostatic pressure in the case of the 50-mm wet bed is nearly 3,000 Pa, whereas for the dry bed water levels have not yet begun to rise at the leeside face. Between seconds 3 and 4, where the most substantial difference in net force appears in Fig. 17, there are quite significant differences between the hydrostatic pressures acting on the leeside face which help explain the reduction in net force. It is thought that the wake closes faster for increasing downstream depths as a result of increasing bore depths, as shown in Fig. 14. After the flow has been split into two streams by the presence of the column, the tendency to converge at the center behind the column is directly influenced by the lateral pressure gradient induced by differences in water levels at the side and center of the channel. Therefore, it should be expected that water levels in the wake will rise the fastest in the case of the 50-mm wet bed condition, which is exactly what the model predicts.

Further reductions in the peak hydrodynamic force in the case of the 50-mm wet bed may be a result of reduced velocities between seconds 3 and 4. As shown in Fig. 12, it is the only case where depth-averaged velocity in the stream-wise direction is still affected by the bed condition. Since dynamic pressure is proportional to the square of velocity, even small reductions in the velocity can have a large influence on the magnitude of the net force.

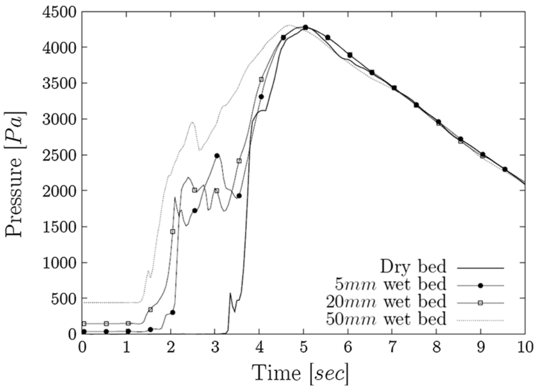


Fig. 18 Time-history of the hydrostatic pressure acting on the leeside face of column for different bed conditions

5 Conclusions

A three-dimensional multiphase numerical model was used to investigate the interaction between a structural model with a square cross-section and tsunami-like bores propagating over dry and wet channel beds of varying depths. The numerical model was first validated by comparing its results to those obtained from a physical model experiment conducted at the NRC-CHC in Ottawa, Canada, for which excellent agreement is observed. The physical model program was developed based on the demonstrated analogy between tsunami-induced bores and dam-break waves generated by the sudden release of an impounded volume of water. Following its validation, the numerical model was further applied to measure the influence of bed condition on propagation characteristics before impact with the structural model. The results showed that even a thin layer of water can substantially increase the steepness and depth of the bore-front. It was shown that further increases in the downstream depth did not significantly alter the shape of the leading edge of the bore, which plays an important role in development of the impulsive pressures at impact. The additional water being pushed by the generated bores for increasing downstream depths tended to build up at a location upstream of the leading edge, forming an undulation on the surface and increasing the inundation depth momentarily after passage of the leading edge.

In the case of bores propagating over a wet channel bed, the bore-front region flows over top of the initially quiescent downstream water until a point located upstream where momentum has transferred uniformly throughout the vertical. As a result, the dynamic pressure generated at impact is elevated above the channel bottom. This creates a divergence point where a portion of the flow is redirected upwards to create the splash and the other portion is diverted downwards which becomes part of a horseshoe vortex that wraps around the sides of the column. The horseshoe vortex also develops in the case of the initially dry channel bed, albeit much later than for the wet channel beds.

Small reductions in the peak hydrodynamic force were observed for the 5- and 20-mm wet beds. However, a much more substantial reduction was observed in the case of the 50-mm initial downstream depth. It was found that water levels in the wake of the structure rose faster for increasing initial downstream depths, reducing the differential hydrostatic pressure between the upstream and downstream faces of the structures. Additionally, sustained reductions in the depth-averaged velocity reduced the dynamic pressure acting on the upstream column face in the case of the 50-mm wet bed.

This study shows that the bed condition is an important factor in the development of hydrodynamic loads on structures in tsunami-prone regions. While entrained air and bore profile shape are dominant factors in generation of impulsive loads at impact, flow velocity and evolution of the wake may govern loading at later stages, all of which are highly dependent on the initial bed condition.

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